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MAGNETIC DISK DEVICE AND METHOD OF MANUFACTURING MAGNETIC DISK DEVICE

Abstract

A magnetic disk device of an embodiment includes: a housing in which a base and a cover are combined and airtightly closed; a magnetic disk housed in the housing; a needle protruding upward in an internal space of the housing; and a desiccant assembly provided on the cover so as to face the base away from the base by a predetermined distance, in which the desiccant assembly includes: a case including an opening on a surface facing the base, a sealing film provided at the opening, a division filter that divides space of the case into a first chamber and a second chamber, the second chamber including the opening, and a desiccant filled in the first chamber, and the second chamber communicates with the internal space of the housing by the needle penetrating the sealing film.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-025102, filed on Feb. 22, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a magnetic disk device and a method of manufacturing the magnetic disk device.

BACKGROUND

[0003] A magnetic disk device includes a housing that houses various components including a magnetic disk. The housing includes, for example, a base and a cover attached to the base. In order to reduce rotation resistance of the magnetic disk, the inside of the housing is filled with He gas or the like.

[0004] In such a magnetic disk device, desiccant may be enclosed for humidity control in a housing. In order to maintain hygroscopicity of the desiccant, exposure of the desiccant to an atmosphere is preferably inhibited to a very short time when the desiccant is enclosed in the housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is an exemplary perspective view illustrating a magnetic disk device according to a first embodiment in an exploded manner;

[0006] FIGS. 2A and 2B are schematic views illustrating one example of a configuration of a desiccant assembly according to the first embodiment;

[0007] FIGS. 3A and 3B are schematic views illustrating one example of a procedure of a method of manufacturing the magnetic disk device according to the first embodiment;

[0008] FIGS. 4A and 4B are schematic views illustrating one example of the procedure of the method of manufacturing the magnetic disk device according to the first embodiment;

[0009] FIGS. 5A and 5B are schematic views illustrating how a desiccant assembly according to Variation 1 of the first embodiment is assembled to a base;

[0010] FIG. 6 is a schematic view illustrating one example of a needle of the magnetic disk device according to Variation 2 of the first embodiment;

[0011] FIGS. 7A to 7C are exemplary perspective views illustrating a magnetic disk device according to a second embodiment in an exploded manner;

[0012] FIGS. 8A to 8Db are schematic views illustrating one example of a configuration of a desiccant assembly according to the second embodiment;

[0013] FIG. 9 is a schematic view illustrating one example of a procedure of a method of manufacturing the magnetic disk device according to the second embodiment;

[0014] FIGS. 10A and 10B are schematic views illustrating one example of the procedure of the method of manufacturing the magnetic disk device according to the second embodiment;

[0015] FIG. 11 is a schematic view illustrating one example of a needle of the magnetic disk device according to Variation 1 of the second embodiment;

[0016] FIGS. 12A and 12B are schematic views illustrating one example of a configuration of the

desiccant assembly according to Variation 2 of the second embodiment; and

[0017] FIGS. **13A** to **13C** are a schematic views illustrating some examples of needles of the magnetic disk device according to Variation 3 of the second embodiment.

DETAILED DESCRIPTION

[0018] A magnetic disk device of an embodiment includes: a housing in which a base and a cover are combined and airtightly closed; a magnetic disk housed in the housing; a needle protruding upward in an internal space of the housing; and a desiccant assembly provided on the cover so as to face the base away from the base by a predetermined distance, in which the desiccant assembly includes: a case including an opening on a surface facing the base, a sealing film provided at the opening, a division filter that divides space of the case into a first chamber and a second chamber, the second chamber including the opening, and desiccant filled in the first chamber, and the second chamber communicates with the internal space of the housing by the needle penetrating the sealing film.

[0019] Embodiments of the present invention will be described in detail below with reference to the drawings. Note that the present invention is not limited by the following embodiments.

Furthermore, components in the following embodiments include those that can be easily assumed by those skilled in the art or those that are substantially the same.

First Embodiment

[0020] A first embodiment will be described in detail below with reference to the drawings.

(Configuration Example of Magnetic Disk Device)

[0021] FIG. **1** is an exemplary perspective view illustrating a magnetic disk device **10** according to the first embodiment in an exploded manner. The magnetic disk device **10** of the first embodiment is configured as, for example, a hard disk drive (HDD). Note, however, that the magnetic disk device **10** of the first embodiment may be another magnetic disk device such as a hybrid HDD.

[0022] Note that, in the present specification, an X direction, a Y direction, and a Z direction are defined for convenience. The X direction, the Y direction, and the Z direction are orthogonal to each other. The X direction is defined as a direction along the width of the magnetic disk device **10**. The Y direction is defined as a direction along the depth of the magnetic disk device **10**. The Z direction is defined as a direction along the thickness of the magnetic disk device **10**, and may also be referred to as a vertical direction in the present specification.

[0023] As illustrated in FIG. **1**, the magnetic disk device **10** includes a plurality of magnetic disks **12**, a spindle motor **13**, a plurality of magnetic heads **14**, a head stack assembly (HSA) **15**, a voice coil motor (VCM) **16**, a ramp load mechanism **17**, a flexible printed circuit board (FPC) **18**, a printed circuit board (PCB) **19**, and a housing **20** that covers these members. The magnetic heads **14** are also referred to as sliders.

[0024] The housing **20** includes a base **21**, an inner cover **22**, and an outer cover **23**. The base **21**, the inner cover **22**, and the outer cover **23** are made of metal such as aluminum alloy and stainless steel. Note, however, that the base **21**, the inner cover **22**, and the outer cover **23** may be made of other materials.

[0025] The base **21** is a bottomed container, and has a bottom wall **211** and a side wall **212**. The bottom wall **211** has a shape of a substantially rectangular plate expanding along an XY plane. The side wall **212** protrudes from an edge of the bottom wall **211** in a direction along the Z direction. A plurality of screw holes **213** is provided in the upper surface of the side wall **212**.

[0026] The inner cover **22** has a shape of a substantially rectangular plate expanding along the XY plane. The inner cover **22** includes a plurality of insertion holes **223** at positions corresponding to the screw holes **213** of the side wall **212** of the base **21**.

[0027] Furthermore, the inner cover **22** includes a plurality of recesses **221** slightly recessed from the upper surface of the inner cover **22**. These recesses **221** are provided in the inner cover **22** inside a portion vertically overlapping the side wall **212** of the base **21**. A vent **222** or the insertion holes **223** penetrating the inner cover **22** in a plate thickness direction are provided at central

portions of these recesses **221**. A screw hole (not illustrated) is provided in a part of a member housed in the base **21**. The insertion holes **223** provided at the centers of the recesses **221** are disposed at positions corresponding to screw holes provided in parts of these members. Seals **225** are attached to the vent **222** and the insertion holes **223** through which screws are tightened to close the vent **222** and the insertion holes **223**.

[0028] Note that FIG. **1** illustrates only the vent **222** among the vent **222** and the insertion holes **223** provided at the centers of these recesses **221**.

[0029] A plurality of screws **224** is inserted into the plurality of insertion holes **223** of the inner cover **22**, and fixed to the plurality of screw holes **213** of the base **21**. The inner cover **22** is thus attached to an end of the side wall **212** of the base **21** so as to overlap the base **21** in the Z direction, and fixed to the members housed in the base **21**. A gasket (not illustrated) is interposed between the inner cover **22** and the base **21**.

[0030] The outer cover **23** is airtightly joined to the end of the side wall **212** of the base **21** by, for example, welding so as to cover the inner cover **22**.

[0031] Such a configuration causes the inside of the housing **20** to be sealed. The housing **20** houses the magnetic disks **12**, the spindle motor **13**, the magnetic heads **14**, the HSA **15**, the VCM **16**, the ramp load mechanism **17**, and the FPC **18** described above. Furthermore, the internal space of the housing **20** is filled with gas having a density lower than that of air, such as He gas, other mixed gases, or the like.

[0032] For example, the magnetic disks **12** are a disk-shaped recording media having recording surfaces on which magnetic recording layers are provided on upper surfaces, lower surfaces, and the like. The magnetic disks **12** in an example of FIG. **1** have a diameter of, for example, 3.5 inches. Note, however, that the magnetic disks **12** are not limited to the example.

[0033] The spindle motor **13** supports and rotates the plurality of magnetic disks **12** in an overlapping manner at predetermined intervals in the Z direction in which the recording surfaces thereof face each other. Here, the internal space of the housing **20** is filled with, for example, He gas as described above. This prevents the magnetic disks **12** from receiving air resistance when the magnetic disks **12** rotate at a high speed, so that rotation resistances of the magnetic disks **12** are reduced.

[0034] The magnetic disk device **10** of the first embodiment configured as described above is also referred to as a He-filled HDD, a He-sealed HDD, or the like. The magnetic disks **12** can be made thinner by filling the housing **20** with He gas or the like and lowering the rotation resistances of the magnetic disks **12**. The capacity of the magnetic disk device **10** can be easily increased by increasing the number of stacked magnetic disks **12**.

[0035] The plurality of magnetic disks **12** is held on a hub of the spindle motor **13** by, for example, a clamp spring. In the example of FIG. **1**, the magnetic disk device **10** includes 10 magnetic disks **12** arranged in the Z direction. Note that the number of magnetic disks **12** is not limited to the example.

[0036] The magnetic heads **14** record and reproduce information on and from the recording surfaces of the magnetic disks **12**. In other words, the magnetic heads **14** read and write information from and to the magnetic disks **12**. The magnetic heads **14** are mounted on the HSA **15**.

[0037] The HSA **15** is rotatably attached to the base **21** at a position separated from the magnetic disks **12** in a direction substantially orthogonal to the Z direction. The VCM **16** rotates the HSA **15** to dispose the magnetic heads **14** at desired positions. The ramp load mechanism **17** holds the magnetic heads **14** moved to the outermost peripheries of the magnetic disks **12** at unload positions separated from the magnetic disks **12**.

[0038] The FPC **18** is supported and reinforced by an FPC plate provided on the bottom wall **211** of the base **21**. One end of the FPC **18** is connected to a flexure included in the HSA **15**. The FPC **18** is electrically connected to the magnetic heads **14** via the flexure of the HSA **15**. The other end of

the FPC **18** is connected to a connector provided on the bottom wall **211** of the base **21**.

[0039] The PCB **19** is attached to the outside of the housing **20**, such as the back surface of the bottom wall **211** of the base **21**. The PCB **19** is mounted with various electronic components such as a relay connector, an interface (I/F) connector, and a controller. The relay connector is connected to the connector of the bottom wall **211** of the base **21**. The I/F connector is connected to a host computer. The controller controls the operation of the magnetic disk device **10**. The relay connector is electrically connected to the FPC **18** via the connector of the bottom wall **211** of the base **21**.

[0040] Note that humidity control in the housing **20** is important for a He-filled HDD such as the magnetic disk device **10** of the first embodiment since gas other than He gas, moisture, and the like cause contamination in the housing **20**. Therefore, a desiccant assembly filled with desiccant is enclosed in the housing **20** in the magnetic disk device **10** of the first embodiment.

(Configuration Example of Desiccant Assembly)

[0041] Next, a configuration example of a desiccant assembly **100** enclosed in the magnetic disk device **10** of the first embodiment will be described with reference to FIGS. 2A and 2B.

[0042] FIGS. 2A and 2B are schematic views illustrating one example of a configuration of the desiccant assembly **100** according to the first embodiment. More specifically, FIG. 2A is a cross-sectional view of the desiccant assembly **100** enclosed in the magnetic disk device **10**. FIG. 2B is a top view illustrating an internal structure of the magnetic disk device **10** in which the desiccant assembly **100** is enclosed.

[0043] As illustrated in FIGS. 2A and 2B, the desiccant assembly **100** of the first embodiment is provided on the inner surface of the inner cover **22** of the housing **20** so as to be housed in an empty space in the housing **20**, such as a region surrounded by the HSA **15**, the VCM **16**, and the FPC **18**.

[0044] The desiccant assembly **100** includes a case **110**, a sealing film **120**, a filter **130**, and desiccant **140**.

[0045] The case **110** is made of resin having a shape such as a rectangular parallelepiped having a space therein. Note, however, that the case **110** may be made of metal. In this case, the case **110** may be made of the same metal as the inner cover **22**, such as aluminum alloy and stainless steel. The case **110** may be integrated with the inner cover **22**.

[0046] The case **110** has an upper surface attached to the inner surface of the inner cover **22**, and has an opening **111** on a side of a lower surface facing the base **21**. The sealing film **120** is fitted into the opening **111**. The sealing film **120** has sealability, and prevents transmission of air, gas such as He, moisture in the gas, and the like. The sealing film **120** is a thin film made of, for example, aluminum.

[0047] The filter **130** is provided in the case **110**. The filter **130** divides the space in the case **110** in a direction along the inner surface of the inner cover **22**. The filter **130** is, for example, a membrane filter made of poly-tetra-fluoro-ethylene (PTFE). Air, gas such as He, moisture in the gas, and the like can be transmitted through the filter **130**. Note that the filter **130** is one example of a division filter that divides the space in the case **110** into two spaces **110r** and **110f**.

[0048] The space **110f** among the two spaces **110r** and **110f** in the case **110** partitioned by the filter **130** is provided on a back side of the space **110r** on a side having the opening **111** into which the sealing film **120** is fitted. The space **110f** is filled with the desiccant **140** capable of adsorbing the moisture and the like in the gas. The other space **110r** among the two spaces **110r** and **110f** in the case **110** partitioned by the filter **130** is maintained to be hollow.

[0049] Here, adsorbent that adsorbs the moisture and the like in the gas includes reversible adsorbent and irreversible adsorbent. The reversible adsorbent can recover hygroscopicity by performing vacuum drying and the like even when the reversible adsorbent adsorbs moisture and the hygroscopicity thereof is reduced. The irreversible adsorbent cannot recover once reduced hygroscopicity by vacuum drying and the like, and has difficulty in reproducing at a temperature of 80° or less, which is an upper limit value of the heat resistance of the magnetic disk device **10**. In

contrast, the irreversible adsorbent can maintain the hygroscopicity even in a low-humidity environment with a humidity of 10% or less in which the reversible adsorbent does not function. [0050] In the magnetic disk device **10** of the first embodiment, irreversible adsorbent having excellent hygroscopicity even in a low-humidity environment although recovery of reduced hygroscopicity is difficult is used as the desiccant **140**. For example, synthetic zeolite such as molecular sieves can be used as the desiccant **140** having irreversibility.

[0051] The space **110f** filled with the desiccant **140** and the space **110r** maintained to be hollow among the two spaces **110r** and **110f** in the case **110** partitioned by the filter **130** are examples of a first chamber and a second chamber, respectively.

[0052] A needle **216** is provided on the bottom wall **211** of the base **21** constituting the housing **20** of the magnetic disk device **10**, and protrudes into the empty space in the housing **20** that houses the desiccant assembly **100**. More specifically, the needle **216** is provided at a position where the needle **216** vertically overlaps the opening **111** provided in the case **110** of the desiccant assembly **100**, and has a protrusion height located above the lower surface of the case **110**, that is, a surface facing the base **21**.

[0053] The needle **216** is made of the same metal as the base **21**, such as aluminum alloy and stainless steel, and is integrated with the base **21**. This enables reduction in the number of components of the magnetic disk device **10**. Note, however, that the needle **216** may be made of a material different from that of the base **21**, such as resin and other metal, and may be attached to the base **21** as a part separate from the base **21**.

[0054] The needle **216** penetrates the sealing film **120** of the desiccant assembly **100**. An upper end of the needle **216** reaches the space **110r** in the case **110** of the desiccant assembly **100**.

[0055] The penetration of the needle **216** as described above breaks a part of the sealing film **120** to generate a gap. The gap causes the space **110r** in the case **110** of the desiccant assembly **100** to communicate with the internal space of the housing **20**. This enables the desiccant **140** filling the case **110** of the desiccant assembly **100** to absorb moisture in the housing **20** via the filter **130**, the space **110r** in the case **110**, and the gap of the sealing film **120**.

[0056] Dust and the like that can be generated from the desiccant **140** can be removed through the filter **130** by exposing the desiccant **140** to an atmosphere in the housing **20** via the filter **130**, which inhibits contamination in the housing **20**.

(Method of Manufacturing Magnetic Disk Device)

[0057] Next, a method of manufacturing the magnetic disk device **10** of the first embodiment will be described with reference to FIGS. 3A to 4B. FIGS. 3A to 4B are schematic views illustrating one example of a procedure of the method of manufacturing the magnetic disk device **10** according to the first embodiment.

[0058] In a process of manufacturing the magnetic disk device **10**, the plurality of magnetic disks **12**, the spindle motor **13**, the magnetic heads **14**, the HSA **15**, the VCM **16**, the ramp load mechanism **17**, and the FPC **18** are first assembled so as to be able to operate as the magnetic disk device **10** in the base **21** in which the needle **216** is provided. That is, the magnetic disks **12**, the spindle motor **13**, the magnetic heads **14**, the HSA **15**, the VCM **16**, the ramp load mechanism **17**, and the FPC **18** are housed so as to be able to operate in the space in the base **21**.

[0059] In contrast, the desiccant assembly **100** is attached to the inner surface of the inner cover **22**. Furthermore, the inner cover **22** is caused to overlap the above-described base **21** with, for example, a gasket (not illustrated) being interposed therebetween. The screws **224** are inserted into the insertion holes **223** of the inner cover **22**, and fixed to the screw holes **213** of the base **21**. Furthermore, the seals **225** is attached to the insertion holes **223** provided in the recesses **221** near the center of the inner cover **22** after the screws are tightened. Furthermore, in the case, the PCB **19** connected to the FPC **18** is also attached to the back surface or the like of the base **21** by these screws **224**.

[0060] As described above, the inner cover **22** is airtightly attached to the base **21**. FIGS. 3A and

3B are cross-sectional views illustrating how the inner cover **22** is attached to the base **21**.

[0061] As illustrated in FIG. **3A**, the inner surface of the inner cover **22** to which the desiccant assembly **100** is attached is caused to face the base **21** that houses the magnetic disks **12** and the like. A position of attachment of the desiccant assembly **100** to the inner cover **22** is provided in the case **110**, and adjusted so that the opening **111** into which the sealing film **120** is fitted vertically overlaps the needle **216** provided on the base **21**.

[0062] Furthermore, in a stage in FIG. **3A**, the sealing film **120** fitted into the opening **111** seals the case **110** of the desiccant assembly **100**, and a dry state in which at least moisture and the like are not contained is maintained.

[0063] As illustrated in FIG. **3B**, the needle **216** provided on the base **21** breaks the sealing film **120** of the opening **111** of the case **110** by causing the inner cover **22** to overlap the base **21**. A gap is generated in a part of the sealing film **120**. Furthermore, the lower surface of the case **110** is located above the bottom wall **211** without being in contact with the bottom wall **211** of the base **21** with the inner cover **22** overlapping the base **21**. A gap is generated between the case **110** and the bottom wall **211** of the base **21**.

[0064] This causes the space **110r** in the case **110** to communicate with the space between the inner cover **22** and the base **21** without the bottom wall **211** of the base **21** closing the opening **111** of the case **110**. Therefore, the desiccant **140** filling the case **110** is exposed to an atmosphere of the internal space formed by the inner cover **22** and the base **21** via the filter **130**, the space **110r** in the case **110**, and the gap of the sealing film **120**.

[0065] The internal space formed by the inner cover **22** and the base **21** is initially filled with air containing moisture and the like. Immediately after the inner cover **22** is tightened to the base **21** by the screws, however, air between the inner cover **22** and the base **21** is sucked from the vent **222** provided in the inner cover **22**. Furthermore, He gas or the like fills the space between the inner cover **22** and the base **21** from the vent **222**.

[0066] Note, however, that the air in the space may be discharged from another hole provided in the inner cover **22** by filling the space between the inner cover **22** and the base **21** with He gas or the like from the vent **222** using the other hole together, for example. Examples of the other hole that can be provided in the inner cover **22** include a dust measuring port (not illustrated).

[0067] After the inner cover **22** and the base **21** are assembled, discharge of air and filling with He gas or the like as described above are quickly performed. This enables progress of adsorption of moisture in air, which is started by the desiccant **140** at the timing when the needle **216** penetrates the sealing film **120**, to be inhibited to a short time, and can inhibit a deterioration in the hygroscopicity of the desiccant **140**.

[0068] As illustrated in FIG. **4A**, the seals **225** are then attached to the vent **222** of the inner cover **22**. As described above, when a seal **225** closes the vent **222**, the internal space formed by the inner cover **22** and the base **21** is sealed in a state of being filled with He gas or the like.

[0069] As illustrated in FIG. **4B**, the outer cover **23** is caused to further overlap the inner cover **22**. The outer cover **23** is welded to the end of the side wall **212** of the base **21**.

[0070] As described above, the magnetic disk device **10** of the first embodiment is manufactured.

[0071] As described above, the desiccant assembly **100** is maintained in a sealed state until immediately before the inner cover **22** is attached to the base **21**. Furthermore, after the needle **216** opens the desiccant assembly **100** at the time when the base **21** and the inner cover **22** are assembled, air in the internal space formed by the base **21** and the inner cover **22** is quickly replaced with He gas or the like. This can inhibit exposure of the desiccant **140** to an atmosphere in the manufacturing process to a short time to obtain the magnetic disk device **10** in which the desiccant **140** maintaining high hygroscopicity is enclosed.

Overview

[0072] In recent years, in large-capacity magnetic disk devices, He-sealed HDDs in which He gas or the like is enclosed in a housing have become the mainstream. In the He-sealed HDDs, humidity

control in the housing for enhancing reliability of the magnetic disk devices is an important issue. Humidity may have an influence more remarkably in an HDD in a heat-assisted magnetic recording (HAMR) type in which information is written by a thermal assist.

[0073] Therefore, in a He-sealed HDD, for example, desiccant is enclosed to control humidity in a housing. If time during which the desiccant is exposed to an atmosphere is prolonged in a process of assembling a magnetic disk device, the desiccant may be enclosed with hygroscopicity thereof being deteriorated, or the desiccant may be saturated before the housing is sealed.

[0074] One of prior arts discloses a method of housing a case containing desiccant in a device with the case being sealed by an aluminum seal and opening the desiccant by making a hole in the aluminum seal in the device via a hole of a cover. In the method of the above prior art, however, a hole making process is required to be added. Unfortunately, the addition of the hole making process prolongs a takt time.

[0075] According to the magnetic disk device **10** of the first embodiment, the desiccant assembly **100** includes the case **110**, the sealing film **120**, the filter **130**, and the desiccant **140**. The case **110** has the opening **111** on a surface facing the base **21**. The sealing film **120** covers the opening **111**. The filter **130** divides the space in the case **110** into the space **110f** and the space **110r** having the opening **111**. The space **110f** is filled with the desiccant **140**. The space **110r** communicates with the internal space of the housing **20** by the needle **216** provided on the base **21** penetrating the sealing film **120**.

[0076] This enables the desiccant **140** maintaining hygroscopicity to be enclosed. Therefore, humidity in the housing **20** can be strictly controlled. Reliability of the magnetic disk device **10** can be improved. In the magnetic disk device **10** of an HAMR type, the reliability can be further improved.

[0077] According to the magnetic disk device **10** of the first embodiment, the filter **130** extends in a direction intersecting the inner surface of the inner cover **22**, and divides the space in the case **110** in the direction along the inner surface of the inner cover **22**. The two spaces **110r** and **110f** obtained by the division are arranged side by side in the direction along the inner surface of the inner cover **22**. The desiccant **140** can absorb moisture in the internal space of the housing **20** via the filter **130**, the space **110r**, and the gap of the sealing film **120** provided in the opening **111** by arranging the two spaces **110r** and **110f** as described above.

[0078] According to the method of manufacturing the magnetic disk device **10** of the first embodiment, the base **21** and the inner cover **22** overlap each other. The needle **216** protruding from the side of the base **21** abuts on the sealing film **120**. At least a part of the sealing film **120** is removed from the opening **111** of the case **110** by the needle **216**. The space **110r** in the case **110** thereby communicates with the internal space of the housing **20**. The desiccant assembly **100** is disposed so as to face the base **21** at a position away from the base **21** by a predetermined distance.

[0079] This enables the desiccant **140** to be sealed immediately before the inner cover **22** and the base **21** are assembled. This can inhibit exposure of the desiccant **140** to an atmosphere to a short time. The magnetic disk device **10** in which the desiccant **140** maintaining hygroscopicity is enclosed can be obtained.

[0080] Furthermore, the desiccant assembly **100** can be easily opened at the same time when the inner cover **22** and the base **21** are assembled. A dedicated tool for opening the desiccant assembly **100** is unnecessary unlike in the case of the above prior art. Furthermore, the takt time can be shortened without increasing the number of manufacturing processes. Forgetting to open the desiccant assembly **100** can be also prevented.

[0081] Furthermore, the desiccant assembly **100** is disposed away from the base **21** by a predetermined distance, which inhibits the bottom wall **211** of the base **21** from closing the opening **111** of the case **110**. After assembling of the housing **20**, the desiccant **140** can collect moisture in the housing **20**.

Variation 1

[0082] Next, a desiccant assembly **100a** of Variation 1 of the first embodiment will be described with reference to FIGS. 5A and 5B. The desiccant assembly **100a** of Variation 1 is different from the desiccant assembly **100** of the above-described first embodiment in the mechanism of opening. [0083] FIGS. 5A and 5B are schematic views illustrating how the desiccant assembly **100a** according to Variation 1 of the first embodiment is assembled to the base **21**. Note that, in the following drawings, the same reference signs are attached to the same configurations as those of the above-described first embodiment, and description thereof may be omitted.

[0084] As illustrated in FIG. 5A, the desiccant assembly **100a** of Variation 1 includes a case **110a** having an inner chamber **113**. The inner chamber **113** of the case **110a** is provided in the space **110r**, and has a shape of box covering the opening **111**.

[0085] That is, the inner chamber **113** has a side wall **113s** and an inner lid **113t**. The side wall **113s** surrounds an outer edge of the opening **111**. The inner lid **113t** is provided at an upper end of the side wall **113s** so as to cover the opening **111**, and serves as an upper surface of the inner chamber **113**. One or more openings **114** are provided in the side wall **113s** of the inner chamber **113**. The openings **114** are provided in the side wall **113s** as described above, so that the internal space of the box-shaped inner chamber **113** communicates with the space **110r** in the case **110a**.

[0086] Furthermore, the inner lid **113t** of the inner chamber **113** is preferably disposed at a height position substantially equal to that of the upper end of the needle **216** that is to protrude into the space **110r** of the case **110a** with the inner cover **22** being assembled to the base **21** or a position slightly higher than that of the upper end of the needle **216**.

[0087] A sealing film **120a** is fitted into the opening **111** of the case **110a**. The sealing film **120a** has sealability, and prevents transmission of air, gas such as He, moisture in the gas, and the like. The sealing film **120a** is a thin film made of a resin, such as silicone rubber and fluororubber.

[0088] In a stage in FIG. 5A, the sealing film **120a** fitted into the opening **111** of the case **110a** seals the case **110a** of the desiccant assembly **100a**, and a dry state in which at least moisture and the like are not contained is maintained.

[0089] As illustrated in FIG. 5B, when the inner cover **22** to which the desiccant assembly **100a** is attached is caused to overlap the base **21**, the needle **216** provided on the base **21** abuts on the sealing film **120a**. As described above, the sealing film **120a** has a relatively tough material such as resin. Thus, the sealing film **120a** is pushed upward by the needle **216** without being broken, and sandwiched between the upper end of the needle **216** and the inner lid **113t** of the inner chamber **113**.

[0090] As described above, the opening **111** of the case **110a** is opened by removing the sealing film **120a** with the needle **216** to cause the space **110r** in the case **110a** to communicate with the space between the inner cover **22** and the base **21**.

[0091] According to the magnetic disk device of Variation 1, the desiccant assembly **100a** includes the case **110a**, the inner lid **113t**, and the sealing film **120a**. The case **110a** has the opening **111** on the surface facing the base **21**. The inner lid **113t** is disposed above the opening **111**. The sealing film **120a** is pressed against a surface facing the opening **111** of the inner lid **113t**. The needle **216** is inserted into the space **110r** in the case **110a** through the opening **111**, and abuts on a lower surface of the inner lid **113t** via the sealing film **120a**. This enables the desiccant **140** maintaining hygroscopicity to be enclosed.

[0092] According to the magnetic disk device of Variation 1, other effects similar to those of the magnetic disk device **10** of the first embodiment described above are obtained.

Variation 2

[0093] Next, a magnetic disk device of Variation 2 of the first embodiment will be described with reference to FIG. 6. The magnetic disk device of Variation 2 includes a needle **216d** having a shape different from that of the needle **216** of the first embodiment described above.

[0094] FIG. 6 is a schematic view illustrating one example of the needle **216d** of the magnetic disk device according to Variation 2 of the first embodiment. Note that, in the following drawings, the

same reference signs are attached to the same configurations as those of the above-described first embodiment, and description thereof may be omitted.

[0095] As illustrated in FIG. 6, a base **21d** of Variation 2 includes the needle **216d** including an extending portion **219ax** and a piercing portion **219tp**. The extending portion **219ax** corresponds to a main body portion of the needle **216d**. The extending portion **219ax** penetrates the sealing film **120**, and extends in the vertical direction of the sealing film **120**. The piercing portion **219tp** is located at an upper end portion of the extending portion **219ax**. The piercing portion **219tp** at least partially has a portion having a transverse section larger than that of the extending portion **219ax**. As in an example in FIG. 6, the piercing portion **219tp** may have a shape with a pointed tip portion, such as a conical shape.

[0096] Such a shape of the needle **216d** of Variation 2 causes the piercing portion **219tp** of the needle **216d** to penetrate the sealing film **120** at the time when the inner cover **22** and the base **21d** are assembled. A larger hole HL is thereby made in the sealing film **120**. Furthermore, after the inner cover **22** and the base **21d** are assembled, the extending portion **219ax** having a smaller cross-sectional area than the piercing portion **219tp** is located in the hole HL provided in the sealing film **120**.

[0097] According to the magnetic disk device of Variation 2, the needle **216d** includes the extending portion **219ax** and the piercing portion **219tp**. The extending portion **219ax** extends through the sealing film **120**. The piercing portion **219tp** is provided at the upper end of the needle **216d**, and has a cross-sectional area at least partially larger than the extending portion **219ax**. This enables the larger hole HL to be provided in the sealing film **120**. Hygroscopic efficiency of the desiccant **140** can be further enhanced by collecting moisture in the housing via the hole HL.

[0098] According to the magnetic disk device of Variation 2, other effects similar to those of the magnetic disk device **10** of the first embodiment described above are obtained.

Second Embodiment

[0099] A second embodiment will be described in detail below with reference to the drawings. A magnetic disk device of the second embodiment includes a desiccant assembly having a configuration different from that of the first embodiment. Note that, in the following drawings, the same reference signs are attached to the same configurations as those of the first embodiment, and description thereof may be omitted.

(Configuration Example of Magnetic Disk Device)

[0100] FIGS. 7A to 7C are exemplary perspective views illustrating a magnetic disk device **30** according to the second embodiment in an exploded manner. More specifically, FIG. 7A illustrates the inside of a base **21e** of the magnetic disk device **30**. FIG. 7B illustrates the inner surface of an inner cover **22** of the magnetic disk device **30**. Furthermore, FIG. 7C is a transparent perspective view of the magnetic disk device **30**.

[0101] As illustrated in FIG. 7A, as in the above-described first embodiment, magnetic disks **12**, a spindle motor **13**, magnetic heads **14**, an HSA **15**, a VCM **16**, a ramp load mechanism **17**, an FPC **18**, and an FPC plate **215** are housed in the base **21e** of the magnetic disk device **30**.

[0102] As described above, the FPC plate **215** is a resin member for supporting and reinforcing the FPC **18**, and is disposed on the bottom wall **211** of the base **21e**. In the second embodiment, a needle **216e** is provided on the upper surface of the FPC plate **215**. The needle **216e** can be easily mounted in a limited space in the base **21e** by providing the needle **216e** on the FPC plate **215**.

[0103] The needle **216e** of the second embodiment is made of resin as in the case of the FPC plate **215**, for example. The needle **216e** is integrated with the FPC plate **215**. This enables reduction in the number of components of the magnetic disk device **30**. Note, however, that the needle **216e** may be made of a material, such as metal, different from that of the FPC plate **215**, and embedded in the FPC plate **215**.

[0104] As illustrated in FIG. 7B, a desiccant assembly **300** of the second embodiment is attached to the inner surface of the inner cover **22** of the magnetic disk device **30**, that is, a surface facing the

base **21e** in a state of being assembled to the base **21e**. More specifically, the desiccant assembly **300** is attached to the inner surface of the inner cover **22** at a position overlapping the vent **222** provided in the inner cover **22**.

[0105] As illustrated in FIG. 7C, the desiccant assembly **300** is disposed at a position where the desiccant assembly **300** vertically overlaps the needle **216e** of the FPC plate **215** installed on the base **21e** with the base **21e** and the inner cover **22** being assembled.

(Configuration Example of Desiccant Assembly)

[0106] Next, a detailed configuration example of the desiccant assembly **300** of the second embodiment will be described with reference to FIGS. 8A to 8Db.

[0107] FIGS. 8A to 8Db are schematic views illustrating one example of a configuration of the desiccant assembly **300** according to the second embodiment. FIGS. 8A to 8Db illustrate a configuration example of the desiccant assembly **300** alone before attachment to the inner cover **22**.

[0108] More specifically, FIG. 8A is a perspective view illustrating the entire configuration of the desiccant assembly **300**. FIG. 8B is a perspective cross-sectional view illustrating the entire configuration of the desiccant assembly **300**. FIGS. 8Ca and 8Cb illustrate a case **310** of the desiccant assembly **300**. FIG. 8Ca is a perspective view of the case **310**. FIG. 8Cb is a top view of the case **310**. FIGS. 8Da and 8Db illustrate a sealing film **320** to be attached to the case **310** of the desiccant assembly **300**. FIG. 8Da is a perspective view on a back surface side of the sealing film **320**. FIG. 8Db is a perspective view on an upper surface side of the sealing film **320**.

[0109] As illustrated in FIG. 8A, the desiccant assembly **300** before attachment includes the case **310**, the sealing film **320**, and a release sheet **360**. The case **310** has, for example, a columnar shape having substantially flat upper and lower surfaces. The sealing film **320** is attached to the lower surface of the case **310**, that is, a surface to face the FPC plate **215** on the base **21e** at the time of attachment to the inner cover **22**. The release sheet **360** is attached to the upper surface of the case **310**, that is, a surface of attachment to the inner cover **22**.

[0110] As illustrated in FIG. 8B, the case **310** has an opening **311** on the lower surface, and has flow paths **312** and vents **313** on the upper surface. A filter **330** is provided in the case **310**. The filter **330** extends in a direction along the upper and lower surfaces of the case **310**, and divides space in the case **310** into two spaces **310r** and **310f** in the vertical direction.

[0111] Note, however, that the case **310** itself may include two cases including the spaces **310r** and **310f**. In this case, these two cases may be integrated as the case **310** including the two spaces **310r** and **310f** by bonding the two cases via the filter **330** or fitting the two cases into each other.

[0112] As in the case of the filter **130** of the first embodiment described above, the filter **330** is a membrane filter made of PTFE, for example. The filter **330** can transmit air, gas such as He, moisture in the gas, and the like. Note that the filter **330** of the second embodiment is one example of a division filter that divides the space in the case **310** into the two spaces **310r** and **310f**.

[0113] The space **310r** among the two spaces **310r** and **310f** in the case **310** partitioned by the filter **330** has the opening **311** which the case **310** has on the lower surface, and is disposed on the lower surface side of the case **310**. The sealing film **320** is attached to the lower surface of the case **310** so as to cover the opening **311** and seal the space **310r** and the like in the case **310**.

[0114] The space **310f** among the two spaces **310r** and **310f** in the case **310** partitioned by the filter **330** includes the vents **313** and the like which the case **310** has on the upper surface, and is disposed on the upper surface side of the case **310**. The release sheet **360** is attached to the upper surface of the case **310** by an annular double-sided tape **350**. The release sheet **360** is a thin film made of, for example, aluminum.

[0115] The double-sided tape **350** has an annular shape as described above. An outer edge of the release sheet **360** is attached to an outer edge of the upper surface of the case **310** around the flow paths **312** and the vents **313**. This causes the release sheet **360** to cover the flow paths **312** and the vents **313** to seal the space **310f** and the like in the case **310**.

[0116] The space **310f** in the case **310** is filled with desiccant **340**. Also in the second embodiment,

for example, irreversible adsorbent including synthetic zeolite such as molecular sieves can be used as the desiccant **340**. Furthermore, the filter **330** of the second embodiment has a function of collecting dust generated from the desiccant **340**. In contrast, the space **310r** in the case **310** is maintained to be hollow.

[0117] The space **310f** filled with the desiccant **340** and the space **310r** maintained to be hollow among the two spaces **310r** and **310f** in the case **310** partitioned by the filter **330** are examples of a first chamber and a second chamber, respectively.

[0118] As illustrated in FIGS. **8Ca** and **8Cb**, a plurality of flow paths **312** on the upper surface of the case **310** extends radially from, for example, a central portion. The vents **313** penetrating the upper surface of the case **310** in the plate thickness direction are connected to ends of the plurality of radially extending flow paths **312**. As described above, the release sheet **360** attached on the upper surface of the case **310** closes these flow paths **312** and the vents **313**.

[0119] As illustrated in FIGS. **8Da** and **8Db**, the sealing film **320** attached to the lower surface of the case **310** is a thin film made of, for example, aluminum.

[0120] A thin portion **321** obtained by further thinning the thin sealing film **320** is provided in a central portion of a back surface side of the sealing film **320**, that is, a surface that is to face the FPC plate **215** on the base **21e** at the time of attachment to the inner cover **22**. As illustrated in FIG. **8Da**, the thin portion **321** has a shape such as a cross shape so as to be easily broken by the needle **216e** of the FPC plate **215**.

[0121] An adhesive layer **322** such as an adhesive is annularly provided on an outer peripheral portion on the upper surface side of the sealing film **320**, that is, a surface on a side to be attached to the lower surface of the case **310**. The sealing film **320** closes the opening **311** to seal the space **310r** and the like in the case **310** by bonding the sealing film **320** on the outer edge of the lower surface of the case **310** around the opening **311** of the case **310** with the adhesive layer **322**.

Furthermore, the annular adhesive layer **322** is disposed around the opening **311** of the case **310**, which does not prevent the needle **216e** of the FPC plate **215** from penetrating the sealing film **320**, and inhibits the adhesive layer **322** from causing contamination after the sealing film **320** is broken.

[0122] As described above, the sealing film **320** and the adhesive layer **322** are configured as, for example, an aluminum seal as a whole.

[0123] The above-described configuration causes the case **310** of the desiccant assembly **300** to be sealed by the sealing film **320** attached to the opening **311** and the release sheet **360** attached to the flow paths **312** and the vents **313** in the desiccant assembly **300** alone before attachment to the inner cover **22**, and maintains a dry state not containing at least moisture and the like.

(Method of Manufacturing Magnetic Disk Device)

[0124] Next, a method of manufacturing the magnetic disk device **30** of the second embodiment will be described with reference to FIGS. **9** to **10B**. FIGS. **9** to **10B** are schematic views illustrating one example of a procedure of the method of manufacturing the magnetic disk device **30** according to the second embodiment.

[0125] Also in the second embodiment, the plurality of magnetic disks **12**, the spindle motor **13**, the magnetic heads **14**, the HSA **15**, the VCM **16**, the ramp load mechanism **17**, and the FPC **18** are assembled so as to be able to operate as the magnetic disk device **10** in the base **21e**. In the case, the needle **216e** is provided, and the FPC plate **215** that supports the FPC **18** is also installed in the base **21e**.

[0126] In contrast, the desiccant assembly **300** is attached to the inner surface of the inner cover **22**.

[0127] That is, the release sheet **360** of the desiccant assembly **300** is released to expose the double-sided tape **350** on the upper surface of the case **310**. In the case, the desiccant **340** in the case **310** is exposed to an atmosphere via the flow paths **312** and the vents **313**. A process of attaching the desiccant assembly **300** to the inner cover **22** is preferably performed immediately before the inner cover **22** is assembled to the base **21e**.

[0128] Furthermore, when the release sheet **360** is released and the desiccant **340** is exposed to an

atmosphere, air spreads radially from the central portion of the case **310** through the flow paths **312**, and reaches the desiccant **340** via the vents **313**. This prevents the air from flowing into the desiccant **340** at once, and inhibits rapid progress of hygroscopic reaction of the desiccant **340**. The rapid hygroscopic reaction of the desiccant **340** may remarkably deteriorate the hygroscopicity of the desiccant **340**.

[0129] The desiccant assembly **300** from which the release sheet **360** has been released is attached to the inner surface of the inner cover **22** by the double-sided tape **350**. In the case, the position of attachment of the desiccant assembly **300** is adjusted such that the central portion on the upper surface of the case **310** overlaps the vent **222** provided in the inner cover **22**.

[0130] Note that the vents **313** provided in the case **310** are examples of second vents. The vent **222** provided in the inner cover **22** and connected to the vents **313** of the case **310** are examples of a first vent.

[0131] Even after the desiccant assembly **300** is attached to the inner surface of the inner cover **22**, the desiccant **340** in the case **310** is continuously exposed to external air via the flow paths **312** and the vents **313** provided in the case **310** and the vent **222** of the inner cover **22**. Therefore, a temporary fixing seal **225p** is attached to the vent **222** of the inner cover **22** to which the desiccant assembly **300** is attached until the inner cover **22** is assembled to the base **21e**.

[0132] As described above, the inner cover **22** to which the desiccant assembly **300** is attached is attached to the base **21e** on which the plurality of magnetic disks **12** and the like are housed and the FPC plate **215** is installed. FIGS. **10A** and **10B** are cross-sectional views illustrating the state.

[0133] As illustrated in FIG. **10A**, the inner surface of the inner cover **22** to which the desiccant assembly **300** is attached is caused to face the base **21e** that houses the magnetic disks **12** and the like. The desiccant assembly **300** is attached at the position of the vent **222** of the inner cover **22**, so that the opening **311** provided in the case **310** and attached to the sealing film **320** is disposed so as to vertically overlap the needle **216e** provided on the FPC plate **215** of the base **21e**.

[0134] Furthermore, in a stage in FIG. **10A**, the case **310** of the desiccant assembly **300** is sealed by the sealing film **320** attached to the opening **311** and the temporary fixing seal **225p** attached to the vent **222** of the inner cover **22**.

[0135] As illustrated in FIG. **10B**, the needle **216e** provided on the FPC plate **215** of the base **21e** breaks the sealing film **320** that closes the opening **311** of the case **310** by causing the inner cover **22** to overlap the base **21e**. A gap is generated in a part of the sealing film **320**. Furthermore, the lower surface of the case **310** is located above the FPC plate **215** with the inner cover **22** overlapping the base **21e** without being in contact with the upper surface of the FPC plate **215**. A gap is generated between the case **310** and the upper surface of the FPC plate **215**.

[0136] This causes the space **310r** in the case **310** to communicate with the space between the inner cover **22** and the base **21e** without the upper surface of the FPC plate **215** closing the opening **311** of the case **310**. Therefore, the desiccant **340** filling the case **310** is exposed to an atmosphere of the internal space formed by the inner cover **22** and the base **21e** via the filter **330**, the space **310r** in the case **310**, and the gap of the sealing film **320**.

[0137] After the inner cover **22** is tightened to the base **21e** by screws, air in the internal space between the inner cover **22** and the base **21e** is quickly replaced with He gas or the like from the vent **222** provided in the inner cover **22**. Also in this case, the space may be filled with He gas or the like after air is discharged. The air may be discharged with filling of the He gas or the like by using another hole such as a dust measuring port together.

[0138] Note that it is preferable that the vent **222** of the inner cover **22** to which the desiccant assembly **300** is connected is exclusively used for injecting He gas or the like into the internal space between the inner cover **22** and the base **21e** and that air is sucked from the internal space through another hole. Thus, the hygroscopic function of the desiccant **340** enables He gas or the like to flow into the internal space between the inner cover **22** and the base **21e** in a drier state.

[0139] Also in the case, the filter **330** of the second embodiment exerts a dust collecting function in

order to prevent dust in the outside air from flowing into the internal space between the inner cover **22** and the base **21e**. Furthermore, the flow paths **312** and the vents **313** provided in the upper surface of the case **310** alleviate the speed of the outside air flowing into the desiccant **340**, so that the rapid progress of the hygroscopic reaction of the desiccant **340** is inhibited.

[0140] After the internal space between the inner cover **22** and the base **21e** is filled with He gas or the like, the vent **222** of the inner cover **22** is closed with the seal **225p** to seal the internal space formed by the inner cover **22** and the base **21e**. The outer cover **23** is caused to further overlap the inner cover **22**. The outer cover **23** is welded to the end of the side wall **212** of the base **21e**.

[0141] As described above, the magnetic disk device **30** of the second embodiment is manufactured.

Overview

[0142] According to the magnetic disk device **30** of the second embodiment, the filter **330** of the desiccant assembly **300** extends in the direction along the inner surface of the inner cover **22**, and divides the space in the case **310** in a direction intersecting the inner surface of the inner cover **22**. The space **310f** in the case **310** is disposed between the space **310r** in the case **310** and the inner cover **22**. The desiccant **340** can absorb moisture in the internal space of the housing via the filter **330**, the space **310r**, and the gap of the sealing film **320** attached to the opening **311** also by arranging the two spaces **310r** and **310f** as described above.

[0143] According to the magnetic disk device **30** of the second embodiment, the inner cover **22** has the vent **222** that penetrates the inner cover **22**. The case **310** of the desiccant assembly **300** has the vents **313** that connect the vent **222** of the inner cover **22** with the space **310f** in the case **310** filled with the desiccant **340**.

[0144] This enables an atmosphere in the internal space between the inner cover **22** and the base **21e** to be replaced via the vent **222** of the inner cover **22** and the vents **313** on the upper surface of the case **310**. In the case, if the vent **222** is used for inflow of He gas or the like, drier He gas or the like can be injected into the internal space between the inner cover **22** and the base **21e**. The vent **222** is provided in the inner cover **22**. The desiccant assembly **300** is connected to the vent **222**.

[0145] According to the magnetic disk device **30** of the second embodiment, the case **310** of the desiccant assembly **300** has the plurality of flow paths **312** radially extending from the position immediately below the vent **222** of the inner cover **22**. The vents **313** on the upper surface of the case **310** are connected to the ends of the plurality of flow paths **312**. This enables the speed of inflow of the outside air flowing into the desiccant assembly **300** to be lowered via the vent **222** of the inner cover **22**, and can inhibit rapid hygroscopic reaction of the desiccant **340** due to the outside air.

[0146] The magnetic disk device **30** of the second embodiment further includes the seal **225p** that airtightly closes the vent **222** to which the desiccant assembly **300** is connected. This enables the housing of the magnetic disk device **30** to be sealed, and inhibits the inside desiccant **340** from being continuously exposed to an atmosphere.

[0147] According to the magnetic disk device **30** of the second embodiment, the needle **216e** is provided on the FPC plate **215** disposed on the base **21e**. This enables a configuration necessary for opening the desiccant assembly **300** to be mounted in the limited space in the base **21e**.

Furthermore, since a region in the base **21e** on which the FPC plate **215** is installed has relatively spare space, a space for housing the desiccant assembly **300** attached to the inner cover **22** is easily secured.

[0148] According to the magnetic disk device **30** of the second embodiment, the sealing film **320** has the thin portion **321** processed to be thinner than other portions at the portion which the needle **216e** penetrates. This enables the needle **216e** to easily break the sealing film **320**.

[0149] Note that the sealing film **120** of the desiccant assembly **100** of the first embodiment described above may also have a thin portion as the sealing film **320** of the second embodiment.

[0150] According to the magnetic disk device **30** of the second embodiment, other effects similar to

those of the magnetic disk device **10** of the first embodiment described above are obtained.

Variation 1

[0151] Next, a magnetic disk device of Variation 1 of the second embodiment will be described with reference to FIG. **11**. The magnetic disk device of Variation 1 has the needle **216** at a portion different from that in the above-described second embodiment.

[0152] FIG. **11** is a schematic view illustrating one example of the needle **216** of the magnetic disk device according to Variation 1 of the second embodiment. Note that, in the following drawings, the same reference signs are attached to the same configurations as those of the above-described second embodiment, and description thereof may be omitted.

[0153] As illustrated in FIG. **11**, the magnetic disk device of Variation 1 has the needle **216** on the base **21** as in the above-described first embodiment. Furthermore, an FPC plate **215a** of Variation 1 has a through hole **15th** through which the needle **216** provided on the base **21** passes.

[0154] According to the magnetic disk device of Variation 1, effects similar to those of the magnetic disk device **30** of the second embodiment described above are obtained also by providing the needle **216** on the base **21**.

[0155] Note that the desiccant assembly **100** of the first embodiment described above may be disposed on the FPC plate that supports the FPC **18**. The needle **216** provided on the base **21** may penetrate the FPC plate **215a** of Variation 1 described above and protrude into the housing **20** as a mechanism of opening the desiccant assembly **100**. This enables further space saving in the magnetic disk device **10** of the first embodiment.

[0156] Furthermore, as in the above-described second embodiment, the needle **216e** provided on the upper surface of the FPC plate **215** can be applied as a mechanism of opening the desiccant assembly **100** of the above-described first embodiment.

Variation 2

[0157] Next, a desiccant assembly **300a** of Variation 2 of the second embodiment will be described with reference to FIGS. **12A** and **12B**. The desiccant assembly **300a** of Variation 2 is different from the desiccant assembly **300** of the second embodiment described above in that the desiccant assembly **300a** includes a filter **315** also at a lower end.

[0158] FIGS. **12A** and **12B** are schematic views illustrating one example of a configuration of the desiccant assembly **300a** according to Variation 2 of the second embodiment. More specifically, FIG. **12A** is a perspective view of the desiccant assembly **300a** of Variation 2 as viewed from below. FIG. **12B** is a cross-sectional view illustrating how the desiccant assembly **300a** of Variation 2 is mounted in the magnetic disk device.

[0159] Note that, in the following drawings, the same reference signs are attached to the same configurations as those of the above-described second embodiment, and description thereof may be omitted.

[0160] As illustrated in FIG. **12A**, the desiccant assembly **300a** of Variation 2 includes the filter **315** on a lower surface of the case **310**. The filter **315** surrounds the outer edge of the lower surface of the case **310**, and has an annular shape protruding from the lower surface of the case **310**. The protrusion height of the filter **315** from the lower surface of the case **310** is substantially the same as the height of a gap generated between the lower surface of the desiccant assembly **300a** and the upper surface of the FPC plate **215** at the time when the desiccant assembly **300a** is mounted in the magnetic disk device, for example.

[0161] As in the case of the filter **330**, the filter **315** is a membrane filter made of PTFE, for example. The filter **315** can transmit air, gas such as He, moisture in the gas, and the like.

[0162] As illustrated in FIG. **12B**, in a state in which the desiccant assembly **300a** of Variation 2 is mounted in the magnetic disk device, the filter **315** protruding from the lower surface of the case **310** is in contact with the upper surface of the FPC plate **215**. The annular filter **315** is in contact with the upper surface of the FPC plate **215**. Thus, the filter **315** shields the space between the lower surface of the case **310** and the upper surface of the FPC plate **215** from another internal

space of the housing.

[0163] Note that the filter **315** of Variation 2 is one example of a shielding filter that shields the space between the case **310** and the base **21e** from the internal space of the housing.

[0164] When the needle **216e** penetrates the sealing film **320** provided on the lower surface of the case **310**, dust C and the like may be generated from the broken sealing film **320**. The above-described filter **315** inhibits the dust C scattered in the space between the lower surface of the case **310** and the upper surface of the FPC plate **215** from scattering into the internal space of the housing. Note, however, that the filter **315** transmits air, gas such as He, moisture in the gas, and the like, so that hygroscopic control in the housing performed by the desiccant **340** is not hindered.

[0165] According to the magnetic disk device of Variation 2, the desiccant assembly **300a** further includes the filter **315** that surrounds an outer peripheral portion of a surface facing the base **21e** of the case **310** and that shields the space between the case **310** and the base **21e** from the internal space of the housing. This can inhibit, for example, the dust C generated from the sealing film **320** broken by the needle **216e** from scattering into the space in the housing.

[0166] According to the magnetic disk device of Variation 2, other effects similar to those of the magnetic disk device **30** of the second embodiment described above are obtained.

Variation 3

[0167] Next, a magnetic disk device of Variation 3 of the second embodiment will be described with reference to FIGS. **13A** to **13C**. The magnetic disk device of Variation 3 has any of needles **216a** to **216c** having a shape different from that in the above-described second embodiment.

[0168] FIGS. **13A** to **13C** are a schematic views illustrating some examples of the needles **216a** to **216c** of the magnetic disk device according to Variation 3 of the second embodiment. Note that, in the following drawings, the same reference signs are attached to the same configurations as those of the above-described second embodiment, and description thereof may be omitted.

[0169] In an example in FIG. **13A**, the needle **216a** has a plurality of recesses **217a** at the upper end. The plurality of recesses **217a** is partitioned from each other by, for example, ridges radially extending to the periphery with a central portion of the upper end as a vertex. Furthermore, the depths of these recesses **217a** are adjusted such that the recesses **217a** are recessed to positions below the sealing film **320** after penetration of the needle **216a**. That is, after the needle **216a** penetrates the sealing film **320**, the bottom surfaces of these recesses **217a** are located below the sealing film **320**.

[0170] This enables an atmosphere in the housing to communicate with the space **310r** in the case **310** also via the recesses **217a** of the upper end of the needle **216a** in addition to the gap of the sealing film **320** generated between the sealing film **320** and the needle **216a**. Furthermore, even when the broken sealing film **320** is wound around the needle **216a** to close the gap, a state of communication between the inside of the housing and the space **310r** in the case **310** is secured via the recesses **217a** at the upper end of the needle **216a**.

[0171] In an example in FIG. **13B**, the needle **216b** has a plurality of grooves **217b** radially extending from the pointed tip to the periphery and extending downward on a side wall of the needle **216b**. The plurality of grooves **217b** extends on the side wall of the needle **216b** at least to positions below the sealing film **320** after penetration of the needle **216b**.

[0172] This enables an atmosphere in the housing to communicate with the space **310r** in the case **310** also via the grooves **217b** of the side wall of the needle **216b** in addition to the gap of the sealing film **320** generated between the sealing film **320** and the needle **216b**. Furthermore, even when the broken sealing film **320** is wound around the needle **216b** to close the gap, a state of communication between the inside of the housing and the space **310r** in the case **310** is secured via the grooves **217b** on the side wall of the needle **216b**.

[0173] In an example in FIG. **13C**, the needle **216c** has a through hole **217c** that penetrates the inside of the needle **216c** in a direction of extension of the needle **216c** from the pointed tip. Furthermore, the FPC plate has a groove **218c** connected to the through hole **217c** in the needle

216c at the lower end of the needle **216c** and extending outward from the lower end of the needle **216c**.

[0174] This enables an atmosphere in the housing to communicate with the space **310r** in the case **310** also via the through hole **217c** penetrating the inside of the needle **216c** and the groove **218c** on the upper surface of the FPC plate in addition to the gap of the sealing film **320** generated between the sealing film **320** and the needle **216c**.

[0175] According to the magnetic disk device of Variation 3, the needle **216a** has, at the upper end, the recesses **217a** recessed to positions below the sealing film **320** which the needle **216a** penetrates.

[0176] Alternatively, the needle **216b** has grooves **217b** extending in the vertical direction of the sealing film **320** which the needle **216b** penetrates.

[0177] Still alternatively, the needle **216c** has the through hole **217c** penetrating the needle **216c** in the direction of extension of the needle **216c**. The FPC plate has, on the side wall, the groove **218c** connected to the through hole **217c** of the needle **216c** and extending outward from the lower end of the needle **216c**.

[0178] This causes a larger gap in the sealing film **320**, which promotes hygroscopic reaction in the housing caused by the desiccant **340**. Furthermore, the recesses **217a** of the needle **216a** or the groove **217b** of the needle **216b** causes the sealing film **320** to be more easily broken.

[0179] In addition, as in Variation 2 of the first embodiment described above, the needle **216d** may be applied as a mechanism of opening the desiccant assembly **300** and the like of the second embodiment described above. The needle **216d** includes the extending portion **219ax** and the piercing portion **219tp**. The extending portion **219ax** extends through the sealing film **320**. The piercing portion **219tp** is provided at the upper end of the needle **216d**, and has a cross-sectional area at least partially larger than the extending portion **219ax**.

[0180] According to the magnetic disk device of Variation 3, other effects similar to those of the magnetic disk device **30** of the second embodiment described above are obtained.

[0181] Note that any of the needles **216a** to **216c** of Variation 3 described above can be applied as the mechanism of opening the desiccant assembly **100** of the first embodiment described above.

[0182] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A magnetic disk device comprising: a housing in which a base and a cover are combined and airtightly closed; a magnetic disk housed in the housing; a needle protruding upward in an internal space of the housing; and a desiccant assembly provided on the cover so as to face the base away from the base by a predetermined distance, wherein the desiccant assembly includes: a case including an opening on a surface facing the base, a sealing film provided at the opening, a division filter that divides space of the case into a first chamber and a second chamber, the second chamber including the opening, and a desiccant filled in the first chamber, and the second chamber communicates with the internal space of the housing by the needle penetrating the sealing film.
2. The magnetic disk device according to claim 1, wherein the division filter extends in a direction intersecting an inner surface of the cover, and divides the space of the case in a direction along the inner surface of the cover, and the first chamber and the second chamber are arranged side by side in the direction along the inner surface of the cover.

3. The magnetic disk device according to claim 1, wherein the division filter extends in a direction along an inner surface of the cover, and divides the space of the case in a direction intersecting the inner surface of the cover, and the first chamber is disposed between the second chamber and the cover.
4. The magnetic disk device according to claim 3, wherein the cover includes a first vent penetrating the cover, and the case includes a second vent connecting the first vent and the first chamber.
5. The magnetic disk device according to claim 4, wherein the case includes a plurality of flow paths radially extending from a position immediately below the first vent, and the second vent includes a plurality of second vents connected to ends of the plurality of flow paths.
6. The magnetic disk device according to claim 4, further comprising a seal that airtightly closes the first vent.
7. The magnetic disk device according to claim 3, wherein the desiccant assembly further includes a shielding filter that surrounds an outer peripheral portion of a surface facing the base of the case and that shields space between the case and the base from the internal space of the housing.
8. The magnetic disk device according to claim 1, wherein the sealing film includes a thin portion processed to be thinner than another portion of the sealing film at a portion which the needle penetrates.
9. The magnetic disk device according to claim 1, wherein the needle is provided on an inner surface of the base.
10. The magnetic disk device according to claim 9, wherein the needle penetrates an FPC plate disposed on the base, and protrudes into the internal space of the housing.
11. The magnetic disk device according to claim 1, wherein the needle is provided on an FPC plate disposed on the base.
12. The magnetic disk device according to claim 11, wherein an upper end of the needle is located in the second chamber, the needle includes a through hole penetrating the needle in a direction of extension of the needle, and the FPC plate includes a groove connected to the through hole of the needle and extending outward from a lower end of the needle.
13. The magnetic disk device according to claim 1, wherein an upper end of the needle is located in the second chamber, and the needle includes, at the upper end, a recess reached to a position below the sealing film which the needle penetrates.
14. The magnetic disk device according to claim 1, wherein an upper end of the needle is located in the second chamber, and the needle includes, at a side wall, a groove extending in upper and lower directions of the sealing film which the needle penetrates.
15. The magnetic disk device according to claim 1, wherein an upper end of the needle is located in the second chamber, and the needle includes: an extending portion that extends through the sealing film, and a piercing portion provided at the upper end of the needle, at least partial portion of the piercing portion having a cross-sectional area larger than the extending portion.
16. The magnetic disk device according to claim 1, wherein the internal space of the housing is filled with gas having a density lower than that of air.
17. A magnetic disk device comprising: a housing in which a base and a cover are combined and airtightly closed; a magnetic disk housed in the housing; a needle protruding upward in an internal space of the housing; and a desiccant assembly provided on the cover so as to face the base away from the base by a predetermined distance, wherein the desiccant assembly includes: a case including an opening on a surface facing the base, an inner lid disposed above the opening, a sealing film pressed against a surface facing the opening of the inner lid, a division filter that divides space of the case into a first chamber and a second chamber, the second chamber including the opening, the first chamber and the second chamber being arranged in a direction along an inner surface of the cover, and a desiccant filled in the first chamber, and the needle is inserted into the second chamber through the opening, and abuts on a lower surface of the inner lid via the sealing

film.

18. A method of manufacturing a magnetic disk device, comprising: installing a magnetic disk in a base in which a needle protrudes; attaching a desiccant assembly on a surface facing the base of a cover including a vent, the desiccant assembly including: a case including an opening on a surface facing the base, a sealing film covering the opening, a division filter that divides space of the case into a first chamber and a second chamber, the second chamber including the opening, and a desiccant filled in the first chamber, causing the cover to which the desiccant assembly is attached to overlap the base on which the magnetic disk is installed and attaching the cover to the base; and injecting gas having a density lower than that of air into an internal space formed by the base and the cover through the vent of the cover and forming a housing in which the base and the cover are combined and airtightly closed, wherein the attaching the cover to the base further includes: communicating the second chamber with the internal space of the housing by causing the base and the cover to overlap each other, causing the needle protruding upward in the internal space between the base and the cover to abut on the sealing film, and removing at least a part of the sealing film from the opening with the needle, and disposing the desiccant assembly at a position away from the base by a predetermined distance so as to face the base.
